

Regional response to light illuminance across the human hypothalamus

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Islay Campbell, Roya Sharifpour, Jose Fermin Balda Aizpurua, Elise Beckers, Ilenia Paparella, Alexandre Berger, Ekaterina Koshmanova, Nasrin Mortazavi, John Read, Mikhail Zubkov, Puneet Talwar, Fabienne Collette, Siya Sherif, Christophe Phillips, Laurent Lamalle, Gilles Vandewalle 

GIGA-CRC Human Imaging, University of Liège, Liège, 4000 Belgium • Faculty of Health, Medicine and Life Sciences, School for Mental Health and Neuroscience, Alzheimer Centre Limburg, Maastricht University, Maastricht, 6229 ER The Netherlands • Institute of Neuroscience (IoNS), Department of Clinical Neuroscience, Université Catholique de Louvain (UCLouvain), Woluwe-Saint-Lambert, 1200 Belgium • Synergia Medical SA, Mont-Saint-Guibert, 1435 Belgium

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Abstract

Light exerts multiple non-image-forming biological effects on physiology including the stimulation of alertness and cognition. However, the subcortical circuitry underlying the stimulating impact of light is not established in humans. We used 7 Tesla functional magnetic resonance imaging to assess the impact of variations in light illuminance on the regional activity of the hypothalamus while healthy young adults (N=26; 16 women; 24.3 ± 2.9 y) were completing two auditory cognitive tasks. We find that, during both the executive and emotional tasks, higher illuminance triggered an activity increase over the posterior part of the hypothalamus, which includes part of the tuberomammillary nucleus and the posterior part of the lateral hypothalamus. In contrast, increasing illuminance evoked a decrease in activity over the anterior and ventral parts of the hypothalamus, encompassing notably the suprachiasmatic nucleus and another part of the tuberomammillary nucleus. Critically, the performance of the executive task was improved under higher illuminance and was negatively correlated with the activity of the posterior hypothalamus area. These findings reveal the distinct local dynamics of different hypothalamus regions that underlie the impact of light on cognition.

eLife assessment

This **fundamental** work describes the complex interplay between light exposure, hypothalamic activity, and cognitive function. The evidence supporting the conclusion is **compelling** with potential therapeutic applications of light modulation. The work will be of broad interest to basic and clinical neuroscientists.

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Introduction

Light exerts multiple non-image-forming (NIF) biological effects that influence the quality of sleep and wakefulness, and higher illuminance is known to stimulate alertness and cognition (Campbell, Sharifpour, et al., 2023). The biological effects of light primarily rely on a subclass of retinal ganglion cells that are intrinsically photosensitive (ipRGCs) because they express the photopigment melanopsin, which is maximally sensitive to photons with wavelength $\sim 480\text{nm}$. IpRGCs combine the light signalling of rods and cones to their intrinsic photosensitivity and, collectively, the biological effects of light present a maximal sensitivity to the shorter blue wavelength of visible light (Tri & Do, 2019 [↗](#)). IpRGCs project to multiple subcortical brain areas and their denser projections are found within the hypothalamus, particularly in nuclei involved in sleep and wakefulness regulation (Scammell et al., 2017 [↗](#); Tri & Do, 2019 [↗](#)). The suprachiasmatic nucleus (SCN), which is the site of the principal circadian clock, receives the strongest inputs from ipRGC inputs, over the anterior part of the hypothalamus (Hattar et al., 2006 [↗](#)). Other nuclei also receive ipRGC projections: the subparaventricular zone, one of the main output routes of the SCN, the ventrolateral preoptic nucleus (VLPO) and the preoptic nucleus (PON) involved in sleep initiation and also found in the anterior part of the hypothalamus; the lateral hypothalamus (LH), site of the orexinergic wake-promoting neurons and melanin-concentrating hormone sleep-promoting neurons, found in contrast over the lateral and posterior parts of the hypothalamus (Scammell et al., 2017 [↗](#); Tri & Do, 2019 [↗](#)).

The brain circuitry underlying the biological effects of light has mostly been uncovered in nocturnal rodent models (Campbell, Sharifpour, et al., 2023; Tri & Do, 2019 [↗](#)). Translation to diurnal human beings, where the later maturation of the cortex allows for complex cognitive processing (Braak & Del Tredici, 2015 [↗](#)), remains scarce. In particular, whether hypothalamus nuclei contribute to the stimulating impact of light on cognition in humans is not established.

We addressed this question using ultra-high-field (UHF) 7 Tesla (7T) functional magnetic resonance imaging (fMRI) in healthy young adults exposed to light of various illuminance while engaged in two different auditory cognitive tasks. We find that higher illuminance increased the activity of the posterior part of the hypothalamus encompassing the mamillary bodies (MB) and parts of the LH and tuberomammillary nucleus (TMN). In contrast, higher illuminance decreased the activity over the anterior and ventral parts of the hypothalamus encapsulating notably the SCN and another part of the TMN. Critically, the pattern of modulation was consistent across the two cognitive tasks. Importantly, the performance of the complex cognitive task was improved under higher illuminance while the activity of the posterior part of the hypothalamus was correlated to task performance. The findings reveal the distinct local dynamics of different hypothalamus areas in response to changing illuminance that may contribute to light's impact on cognition.

Results

Twenty-six healthy young adults (16 women; 24.3 ± 2.9 y; **Suppl. Table 1**) completed two auditory cognitive tasks encompassing, respectively, the executive (Collette et al., 2005 [↗](#)) and emotional (Grandjean et al., 2005 [↗](#)) domains, while alternatively maintained in darkness or exposed to short periods (< 1 min) of light of four different illuminances (0.16, 37, 92, 190 melanopic equivalent daylight illuminance - mel EDI- lux; **Suppl. Table 2**) [Fig.1 [↗](#)]. The hypothalamus of each participant was segmented into 5 subparts – inferior-anterior, superior-anterior, inferior-tubular, superior-tubular, and posterior [Fig.2 [↗](#) A] – so we could consistently extract the regional effect of illuminance change on fMRI blood-oxygen-level-dependent (BOLD) signal over most of the hypothalamus volume.

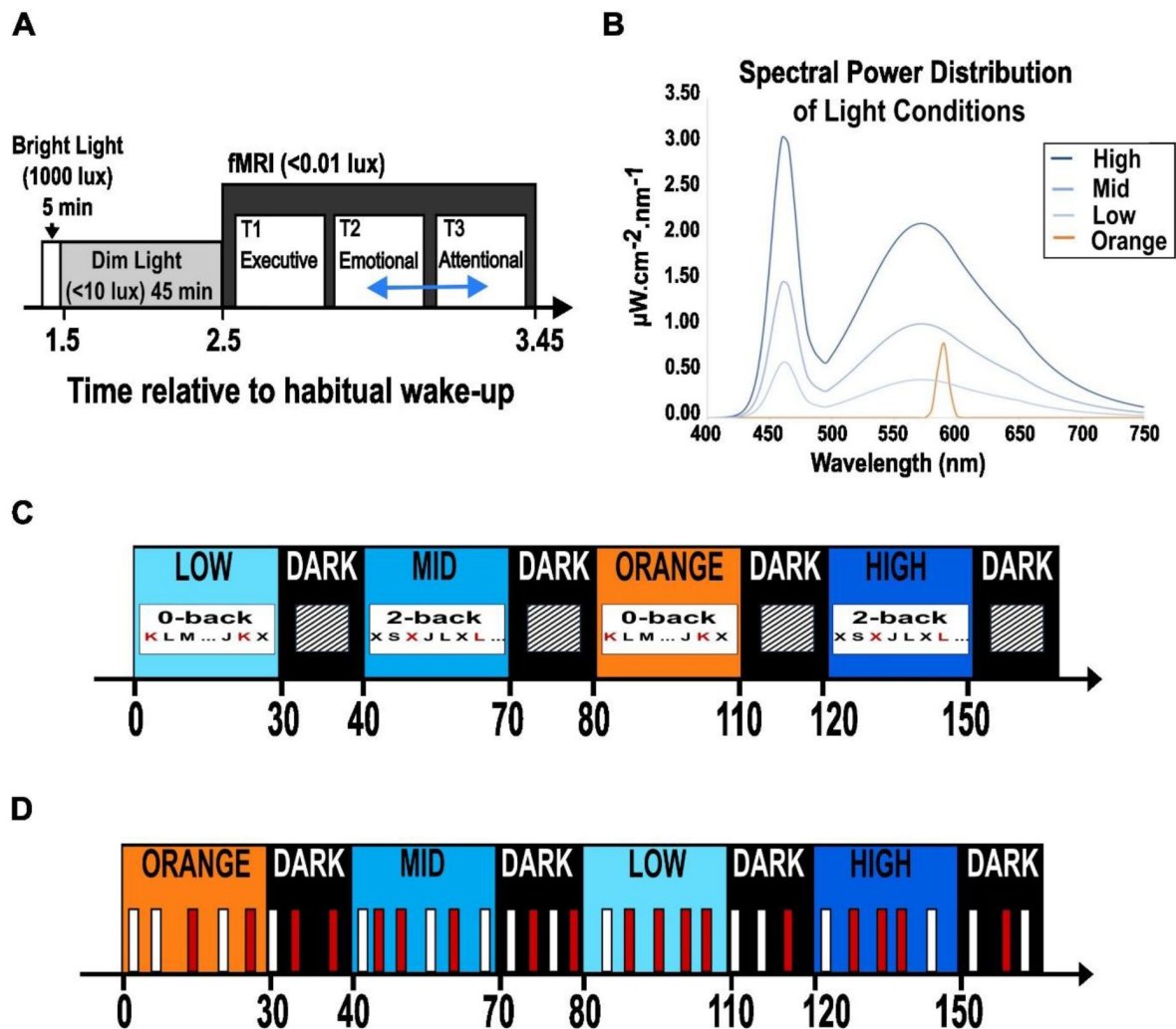


Figure 1.

Experimental protocol.

(A) Overall timeline. After prior light history standardisation, participants performed executive (always first), emotional and attentional tasks (pseudo-randomly 2nd or 3rd, blue arrow). As the attentional task included fewer light conditions, it is not considered in the present manuscript (see methods for more details).

(B) Spectral power distribution of light exposures. Monochromatic orange: 0.16 mel EDI lux; Polychromatic, blue-enriched light (6500K); LOW, MID, HIGH: 37, 92, 190 mel EDI lux. For the present analyses, we discarded colour differences between the light conditions and only considered illuminance as indexed by mel EDI lux, constituting a limitation of our study. See **Suppl. Table 2** for full details).

(C-D) Tasks procedures. Time is reported in seconds relative to session onset; participants were pseudo-randomly exposed to the 4 light conditions. **(C)** Executive task: alternation of letter detection blocks (0-back) and working memory blocks (2-back). **(D)** Emotional task: lure gender discrimination of vocalizations (50% angry (red), 50% neutral (white)).

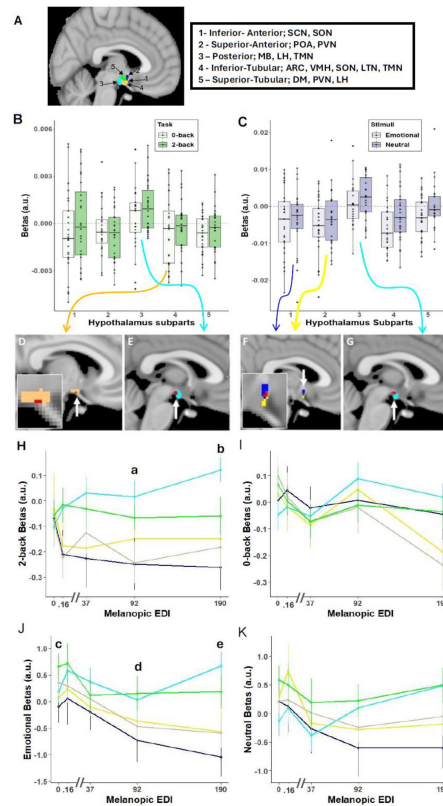


Figure 2.

Illuminance impact on the hypothalamus subparts.

(A) Segmentation of the hypothalamus in five subparts in a representative participant. The nuclei encompassed by the different subparts are indicated in the right inset – according to (Billot et al., 2020). ARC: arcuate nucleus; DMH; dorsomedial nucleus; LH lateral hypothalamus; LTN: lateral tubular nucleus; MB: mamillary body; POA: preoptic area; PVN: paraventricular nucleus; PNH: posterior nucleus of the hypothalamus; SCN: suprachiasmatic nucleus; SON: supraoptic nucleus; TMN: tuberomammillary nucleus; VMN: ventromedial nucleus.

(B-C) Estimates (beta; arbitrary unit – a.u.) of the collective impact of illuminance variation on the activity of each hypothalamus subpart (Refer to [Table 1](#) full statistics). **(B)** Executive task: significant main effect of hypothalamus subparts ($p=0.002$), no significant main of task type ($p=0.4$) or subpart-by-task-type interaction ($p=0.61$). **(C)** Emotional task: significant main effect of hypothalamus subparts ($p<.0001$), and of stimulus type ($p=0.048$) or subpart-by-stimulus-type interaction ($p=0.74$).

(D-G) Whole brain analyses of the collective impact of the variations in illuminance over the hypothalamus area - for illustration. A local positive peak (red; $p_{\text{uncorrected}}<0.001$) was detected over the posterior hypothalamus subpart (light blue) in executive (**E**) and emotional (**G**). A local negative peak (red; $p_{\text{uncorrected}}<0.001$) was detected over the inferior-tubular hypothalamus subparts (light orange) during the executive task (**D**), while local negative peak (red; $p_{\text{uncorrected}}<0.001$) was detected over the inferior-anterior (yellow) and superior-anterior (blue) hypothalamus subparts during the emotional task (**F**) – insets correspond to enlargements over the hypothalamus area. Arrows from panels B and C arise from and are colour coded according to the hypothalamus subpart that is displayed in panels D to G. These results indicate that our finding does not arise from a nearby “leaking” activation/deactivation.

(H-K) Estimates of the impact of each illuminance on the activity of the hypothalamus subparts. (Refer for [Table 2](#) and Suppl. Tables 3/4/5/6 for full statistics) Activity dynamics across illuminance for each subpart (colour code as in A). Results are displayed per task or stimulus type although no interactions with task or stimulus type were detected. Significant illuminance-by-hypothalamus-subpart interactions were detected for **(H-I)** the executive task ($p=0.041$) and **(J-K)** the emotional task ($p=0.041$). Small letter indicate significant difference ($p < 0.05$) between the following subparts at illuminance: a. 92 mel EDI lux: posterior vs. superior-anterior & inferior-tubular; b. 190 mel EDI lux: posterior vs. inferior-anterior, superior-anterior & inferior-tubular; c. 0 mel EDI lux: posterior vs. superior-tubular; d. 92 mel EDI lux: posterior vs. superior-anterior; superior-anterior vs. superior-tubular; e. 190 mel EDI lux: posterior vs. inferior-anterior, superior-anterior & inferior-tubular; superior-tubular vs. superior-anterior, inferior-tubular & inferior-anterior.

The impact of illuminance variations on the activity of the hypothalamus is not uniform

The main analyses aimed at isolating differences in the overall impact of illuminance changes among the 5 hypothalamus subparts. For each subpart, we extracted an index of the illuminance impact as their average regression coefficients between their responses to the tasks and the illuminance levels. These analyses showed significant differences between the hypothalamus subparts for the executive (generalized linear mixed models (GLMM); main effect of the subparts; $p = 0.002$) and emotional (GLMM; main effect of the subparts; $p < 0.0001$) tasks, revealing that, during both tasks, the variations in illuminance affected the activity of the 5 hypothalamus subparts differently [Fig.2 B, C; Table 1]. A nominal main effect of the task was detected for the emotional task [$p = 0.049$; Table 1] but not for the n-back task. For both tasks, there was no significant main effect for any of the other covariates and post hoc analyses showed that the index of the illuminance impact was consistently different in the posterior hypothalamus subpart compared to the other subparts [$p_{\text{corrected}} \leq 0.05$, Table 1]. Importantly, whole-brain analyses confirmed that increasing illuminance resulted in a local increase and decrease of activity that could be detected, respectively, over the posterior and inferior subparts of the hypothalamus [Fig.2 D-G]. This shows that our results do not come from a relatively unspecific and widespread increase in BOLD signal surrounding the hypothalamus subparts and that the effect of light was most prominent over the posterior and inferior-anterior subparts.

Opposite dynamics between the posterior and inferior/anterior hypothalamus at higher illuminance

This prompted us to assess the activity of the hypothalamus subparts under each illuminance to detail the different regional activity dynamics across the hypothalamus. The statistical analyses confirmed that the activity dynamics across illuminance levels differed between the 5 subparts during the executive and the emotional tasks (GLMM; subparts-by-illuminance interaction; $p = 0.041$) tasks [Fig.2 H-K; Table 2]. Post hoc contrasts first considered the impact of the changes in illuminance within each subpart (Suppl. Tables 3, 4). The activity of the posterior hypothalamus subpart significantly ($p < 0.05$) increased under the highest illuminance (190 mel EDI) compared with darkness for both tasks and with the lower illuminances (37 and 92 mel EDI lux) for the emotional task. In contrast, for both tasks, the activity in the inferior-anterior and inferior-tubular hypothalamus subparts significantly ($p < 0.05$) decreased under the highest illuminance compared with darkness, and with lower illuminances for the emotional task. Finally, the activity of the superior anterior hypothalamus subpart decreased under higher illuminance during the emotional but not the executive task, while the activity of the fifth hypothalamus subpart, the superior tubular subpart, was not significantly affected by illuminance changes in either task.

Post hoc analyses also yielded several significant differences between hypothalamus subparts ($p < 0.05$) (Table 2; Suppl. Tables 5, 6). For both tasks, the activity of the posterior hypothalamus subpart was consistently significantly higher than the activity inferior-tubular subpart under the highest illuminances (92 and 190 mel EDI lux). For the executive task, the activity of the posterior hypothalamus subpart was also significantly higher than the superior-anterior subpart under the highest illuminances (92 and 190 mel EDI lux). For the emotional task, the activity of the posterior hypothalamus subpart was also significantly higher than the superior-anterior subpart under the highest illuminances (190 mel EDI lux), while the activity superior-tubular hypothalamus subpart was significantly higher than the activity of the inferior-tubular, inferior-anterior and superior-anterior hypothalamus subparts (92 and/or 190 mel EDI lux). The overall picture arising from these

Executive task							
Main GLMM				Pairwise comparisons			
Effect	F value (df)	P value*	Partial R ²	Contrast [#]	t-value	P _{uncorrected}	P _{corrected}
Hypothalamus subparts	4.36 (4,225)	0.002	0.08	1 vs. 2	-0.30	0.76	0.99
				1 vs. 3	-3.48	0.0006	0.0056
				1 vs. 4	< 0.01	0.99	1
Task	0.74 (1,25)	0.4		1 vs. 5	-0.57	0.57	0.98
Hypothalamus subparts x task type	0.68 (4,225)	0.61		2 vs. 3	-3.17	0.0017	0.015
				2 vs. 4	0.31	0.76	0.99
				2 vs. 5	-0.27	0.79	0.99
Age	0.33 (1,22)	0.57		3 vs. 4	3.48	0.0006	0.0055
BMI	0.59 (1,22)	0.45		3 vs. 5	2.54	0.0041	0.033
Sex	0.01 (1,22)	0.91		4 vs. 5	-0.5	0.57	0.98
Emotional task							
Main GLMM				Pairwise comparisons			
Effect	F Value	P value*	Partial R ²	Contrast [#]	t-value	P _{uncorrected}	P _{corrected}
Hypothalamus subparts	9.38 (4,194)	<.0001	0.22	1 vs. 2	0.67	0.67	0.99
				1 vs. 3	-4.76	< 0.0001	< 0.0001
				1 vs. 4	0.00	0.99	1
Task	4.33 (1,25)	0.048	0.15	1 vs. 5	-1.88	0.06	0.33
Hypothalamus subparts x stimulus type	0.5 (4,194)	0.74		2 vs. 3	-5.43	< 0.0001	0.0001
				2 vs. 4	-0.66	0.51	0.96
				2 vs. 5	-2.54	0.012	0.086
Age	0.43 (1,22)	0.52		3 vs. 4	4.76	< 0.0001	< 0.0001
BMI	0.05 (1,22)	0.83		3 vs. 5	2.85	0.0048	0.038
Sex	1.47 (1,22)	0.24		4 vs. 5	-1.88	0.061	0.32

Table 1.

Differences between hypothalamus subparts in the collective impact of the variation in illuminance on their activity.

Outputs of the generalized linear mixed model (GLMM) with subject as the random factor (intercept and slope), and task and subpart as repeated measures (ar(1) autocorrelation).

* The corrected p-value for multiple comparisons over 2 tests is $p < 0.025$. # Refer to [Fig.2A](#) for correspondence of subpart numbers

Executive task							
Main GLMM				Comparisons between subparts per illuminance #			
Effect	F-value (df)	P value	Partial R ²	Illuminance*	contrast	t-value	p-value
Subpart	1.4 (4,228)	0.23		92	2 vs. 3	-2.25	0.025
Illuminance	2.15 (4,1017)	0.073		92	3 vs. 4	2.58	0.01
				190	1 vs. 3	-2.80	0.0053
Task	3.24 (1,228)	0.073		190	2 vs. 3	-2.24	0.025
Subpart x Illuminance	1.7 (16,1017)	0.041	0.09	190	3 vs. 4	3.15	0.0017
Age	1.19 (1,22)	0.29					
BMI	0.01 (1,22)	0.9					
Sex	0.38 (1,22)	0.54					
Emotional task							
Main GLMM				Comparisons between subparts per illuminance #			
Effect	F-value (df)	P value	Partial R ²	Illuminance*	contrast	t-value	p-value
Subpart	4.29 (4,229)	0.0023	0.07	0	3 vs. 5	-2.05	0.04
				92	2 vs. 3	-2.53	0.012
Illuminance	9.41 (4,1020)	< 0.0001	0.035	92	2 vs. 5	-2.96	0.0032
				190	1 vs. 3	-3.31	0.001
				190	2 vs. 3	-4.75	< 0.0001
Task	0.13 (1,229)	0.72		190	1 vs. 5	-2.5	0.013
				190	2 vs. 5	-4.04	< 0.0001
				190	3 vs. 4	3.13	0.0018
Subpart x Illuminance	1.7 (16,1020)	0.041	0.026	190	4 vs. 5	-2.32	0.021
Age	0.59 (1,22)	0.45					
BMI	1.54 (1,22)	0.23					
Sex	0.05 (1,22)	0.83					

Table 2.

Statistical outputs of GLMM testing for differences between the activity of each subpart of the hypothalamus under each illuminance

* illuminance in mel EDI lux

Only significant comparisons are reported in the main text. For the full table, including post hocs comparing light levels within a subpart, refer to **Suppl. Table 3/4/5/6/7**

comparisons is that higher illuminance increased the activity of the posterior and superior hypothalamus subparts while it decreased the activity of the inferior and anterior hypothalamus subparts.

Performance to the executive task is improved by light and related to the activity of the posterior hypothalamus

Following these analyses, we explored whether the changes in activity across illuminances were related to cognitive performance. We first considered the more difficult (2-back) subtask of the executive task as it requires higher cognitive functions (see method for a full rationale) (Collette et al., 2005 [↗](#)). The analysis revealed that accuracy to the executive task was high in all participants, but accuracy to the more difficult subtask (2-back) improved with increasing illuminance (GLMM; main effect of illuminance; $F = 2.72$; $p = 0.034$; Partial $R^2_{215} = 0.1$; **Fig.3 [↗](#) A**), controlling for age, sex and BMI. Critically, the analysis also showed that performance under each illuminance was significantly related to the activity of the posterior hypothalamus subpart (GLMM; main effect of posterior subpart activity; $F = 9.43$; $p = 0.0027$; Partial $R^2_{219} = 0.09$). Surprisingly, the association was negative (**Fig.3 [↗](#) B**), suggesting that the part of variance explained by the hypothalamus subpart is distinct from the impact of light on performance. In contrast, no significant association was found when considering the activity of the other four subparts (GLMM; main effect of posterior subpart activity; $F < 0.62$; $p > 0.4$; **Fig.3 [↗](#) C,D; Suppl. Table 7**). We went on and found that the accuracy to the simpler control subtask of the executive tasks (0-back, see method) was not associated with the activity of the posterior hypothalamus subpart (GLMM controlling for age, sex and BMI; main effect of subpart activity; $F = 0.57$; $p = 0.45$; **Fig.3 [↗](#) E**), suggesting that the association with performance is specific to the 2-back subtask.

In the last step, we explored the reaction times during the emotional task (accuracy to the lure task is not meaningful). We found that reaction times to the emotional stimuli were not significantly affected by illuminance (GLMM controlling for age, sex and BMI; main effect of illuminance; $F = 1.01$; $p = 0.41$; **Fig.3 [↗](#) F**) and yet, they were significantly associated with the activity of the posterior hypothalamus subpart across each illuminance (GLMM; main effect of subpart activity; $F = 4.34$; $p = 0.04$; **Fig.3 [↗](#) G**). The association was positive meaning that reaction times were longer if activity estimates were higher, which could indicate a reinforcement of the emotional response characterized by longer reaction times (Grandjean et al., 2005 [↗](#)). No such significant association was detected when considering reaction times to the neutral items of the task (GLMM controlling for age, sex and BMI; main effect of illuminance; $F = 1.5$; $p = 0.21$; main effect of subpart activity; $F = 0.28$; $p = 0.6$; **Fig.3 [↗](#) H**).

Discussion

Animal research has established that the biological impact of light illuminance impinges on many subcortical structures, many of which regulate sleep and wakefulness (Campbell, Sharifpour, et al., 2023; Hattar et al., 2006 [↗](#); Scammell et al., 2017 [↗](#); Tri & Do, 2019 [↗](#)). How these findings translate to human beings is not established. Here, we took advantage of the relatively high resolution and signal-to-noise ratio of UHF 7T fMRI to determine how illuminance affects the activity of the hypothalamus as, based on animal research, it receives the densest projections from ipRGCs (Hattar et al., 2006 [↗](#); Tri & Do, 2019 [↗](#)). We find that the activity of the posterior part of the hypothalamus increases with increasing illuminance while, in contrast, the inferior and anterior hypothalamus show a seemingly opposite pattern and see their activity decrease under higher illuminance. The pattern of light-induced changes was consistent across an executive and an emotional task which consisted of block and an event-related fMRI design, respectively. This suggests that a robust anterior-posterior gradient of activity modulation by illuminance is present in hypothalamus across cognitive domains. Importantly, performance to the complex cognitive task was improved under higher illuminance and was correlated to the activity of the posterior

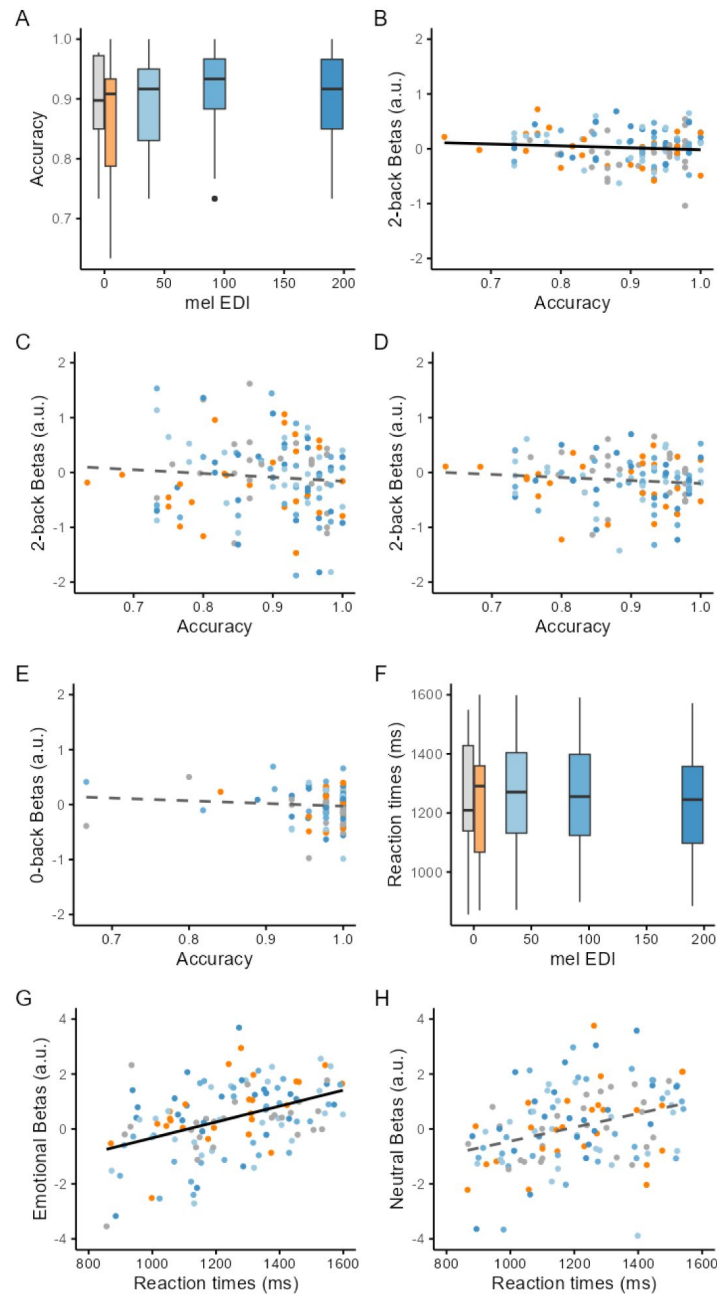


Figure 3.

Impact of illuminance on performance and relationships with the activity of the posterior hypothalamus subpart.

(A) Accuracy (percentage of correct responses) to the 2-back increased with increasing illuminance ($p = 0.034$). (B) Accuracy to the 2-back task is negatively correlated to the activity of the posterior hypothalamus subpart ($p = 0.0027$). (C-D) Accuracy to the 2-back task is not correlated to the activity of the inferior-anterior (C) and inferior-tubular (D) hypothalamus subparts ($p > 0.4$). Association between superior-anterior and superior-tubular subparts are not displayed but were not significant ($p > 0.6$). See Suppl Table 7 for full details. (E) Accuracy to the 0-back task is not correlated to the activity of the posterior hypothalamus subpart ($p = 0.45$). (F) Reaction times to the emotional stimuli did not significantly change with increasing illuminance ($p = 0.41$). (G) Reaction times to the emotional stimuli are correlated to the activity of the posterior hypothalamus subpart ($p = 0.04$) with higher activity associated to slower reaction times. (H) Reaction times to the neutral stimuli are not correlated to the activity of the posterior hypothalamus subpart ($p = 0.6$). Solid and dashed lines correspond to the significant and not significant linear regression lines, respectively.

part of the hypothalamus under the different illuminance, though negatively. The results demonstrate that the human hypothalamus does not respond uniformly to variations in illuminance while engaged in a cognitive challenge and suggest that the posterior part of the hypothalamus may be key in mediating the stimulating impact of light on cognition.

The different nuclei of the hypothalamus do not have clear contrast boundaries based on MRI signals (Billot et al., 2020 [↗](#)). As a result, achieving nucleus resolution over the human hypothalamus even using UFH MRI remains out of reach (Sharifpour et al., 2022 [↗](#)). Therefore, we cannot assign the effects we report to a specific nucleus. We can only speculate and present a selection of plausible scenarios that would need to be tested. The posterior part of the hypothalamus - delineated in each participant based on an automatic reproducible procedure (Billot et al., 2020 [↗](#)) - encompasses the MB as well as parts of the LH and the TMN. All these nuclei could participate to the increased BOLD signal we detect under higher illuminance. The LH and TMN, respectively, produce orexin and histamine, which are both known to promote wakefulness, while animal histology reports direct projection of the ipRGCs to the LH (Hattar et al., 2006 [↗](#); Scammell et al., 2017 [↗](#); Tri & Do, 2019 [↗](#)). Orexin is a good candidate to constitute the circadian signal that promotes wakefulness and to counter the progressive increase in sleep needs with prolonged wakefulness (Zeitler, 2013 [↗](#)). Our data may therefore be compatible with an increase in orexin release by the LH with increasing illuminance. In line with this assumption, chemoactivation of ipRGCs lead to increase c-fos production, a marker of cellular activation, over several nuclei of the hypothalamus, including the lateral hypothalamus (Milosavljevic et al., 2016 [↗](#)). If this initial effect of light we observe over the posterior part of the hypothalamus was maintained over a longer period of exposure, this would stimulate cognition and maintain or increase alertness (Campbell, Sharifpour, et al., 2023) and may also be part of the mechanisms through which daytime light increases the amplitude in circadian variations of several physiological features (Bano-Otalora et al., 2021 [↗](#); Dijk et al., 2012 [↗](#)). It could then also be part of the mechanisms through which evening light may disturb subsequent sleep (Chellappa et al., 2013 [↗](#)) when illuminance is higher than the recommended maximum of 10 mel EDI lux for evening light (Brown et al., 2022 [↗](#)). If the TMN was the hypothalamus nucleus underlying the regional increase in the BOLD signal we report, it could confer a role to histamine in mediating the stimulating impact of light. Of interest, the TMN receives orexin signal from the LH (Scammell et al., 2019 [↗](#)). Alternatively, our findings may suggest a role for the MBs in mediating the impact of light on ongoing cognition, potentially influenced through its innervation by the TMN (Vann, 2010 [↗](#)).

Previous research indicated that increasing illuminance reduced the activity of the anterior part of the hypothalamus encompassing the SCN, either following the exposure to light (Perrin et al., 2004 [↗](#)) or during the exposure (Schoonderwoerd et al., 2022 [↗](#)). We extend this finding by showing that the significant decrease in activity extends beyond the inferior anterior hypothalamus and therefore much beyond the SCN. Chemoactivation of ipRGCs in rodents led to an increase activity of the SCN, over the inferior anterior hypothalamus, but had no impact on the activity of the VLPO, over the superior anterior hypothalamus (Milosavljevic et al., 2016 [↗](#)). How our findings fit with these fine-grained observations and whether there are species-specific differences in the responses to light over the different part of the hypothalamus remains to be established. The inferior-tubular and inferior-anterior subparts of the hypothalamus – that we isolated also based on an automatic reproducible procedure (Billot et al., 2020 [↗](#)) - encompass several nuclei such as notably the SCN, SON, ventromedial nucleus of the hypothalamus, arcuate nucleus and part of the TMN. Again, all these nuclei may be involved in the reduction in BOLD signal we observe at higher illuminance. In terms of chemical communication, these changes in activity could be the results of an inhibitory signal from a subclass of ipRGCs, potentially through the release aminobutyric acid (GABA), as a rodent study found that a subset of ipRGCs release GABA at brain targets including the SCN (and intergeniculate leaflet and ventral lateral geniculate nucleus), leading to a reduction in the ability of light to affect pupil size and circadian photoentrainment (Sonoda et al., 2020 [↗](#)). Whatever the signalling of ipRGC, our finding over the

anterior hypothalamus could correspond to a modification of GABA signalling of the SCN which has been reported to have excitatory properties, such that the BOLD signal changes we report may correspond to a reduction in excitation arising in part from the SCN (Albers et al., 2017 [DOI](#)). Likewise, the SCN is also producing other neuropeptides that could affect its downstream targets. As the inferior-tubular subpart of the hypothalamus also includes part of the TMN it may be the TMN and its GABA production that is decreased by higher illuminance (Scammell et al., 2019 [DOI](#)). The decrease in BOLD signal with increasing illuminance we report could therefore arguably reflect a decreased inhibitory signal arising from the anterior and inferior nuclei of the hypothalamus.

Importantly, none of the scenarios we elaborated on are mutually exclusive and we may have overlooked the potential implication of several nuclei as well as the cellular diversity of the nuclei of the hypothalamus (Adamantidis et al., 2019 [DOI](#); Scammell et al., 2017 [DOI](#)). We further note that the anterior-superior hypothalamus subpart of the hypothalamus encompassing the VLPO and PON sees its activity decreasing under higher illuminance during the emotional task, similarly to the anterior-inferior and inferior-tubular areas. Likewise, similar to the posterior subpart, the activity of the superior-tubular hypothalamus subpart may be increased under higher illuminance during the emotional task. Whether this represents a task-specific effect arising, for instance, from differences in the salience of the auditory stimulus, remains to be determined.

Subcortical structures, and particularly those receiving direct retinal projections, including those of the hypothalamus, are likely to receive light illuminance signal first before passing on the light modulation to the cortical regions involved in the ongoing cognitive process (Campbell, Sharifpour, et al., 2023). A critical aspect of our results is that the performance to the 2-back (executive) subtask was significantly increased when exposed to higher illuminance light. The extent of this increase was limited, likely because performance was overall high at all illuminances, but was not detected for the simpler detection letter subtask (0-back). The result contrasts with many previous 3T MRI investigations on the biological effects of light on human brain function which did not report behavioural changes induced by repeated short exposures to light (e.g. (Vandewalle, Gais, et al., 2007; Vandewalle, Schmidt, et al., 2007; Vandewalle, Archer, et al., 2011) but see (Daneault et al., 2018 [DOI](#); Vandewalle et al., 2006 [DOI](#))). Our 7T MRI study, which includes a sample size larger than many of these previous studies, supports that BOLD fMRI is sensitive in detecting subtle impacts of light on the brain and that these detected changes can arguably contribute to the behavioural changes others reported using longer light exposure and other approaches (e.g. (Cajochen et al., 2011 [DOI](#); Lockley et al., 2006 [DOI](#)) but see (Smolders et al., 2018 [DOI](#))).

Importantly, we find that activity of the posterior hypothalamus subpart is negatively related to the performance to the executive task, making it unlikely that it mediates directly the positive impact of light on performance. The activity of the posterior hypothalamus was, however, associated with an increased behavioural response to emotional stimuli. The association between behaviour and the posterior hypothalamus is therefore likely to be complex and may depend on the context, with for instance different nuclei or neuronal populations contributing in some instances but not in others (Adamantidis et al., 2019 [DOI](#); Scammell et al., 2019 [DOI](#)). It is likely also to work jointly with the decreased activity of the anterior/inferior hypothalamus we detected as well as with other non-hypothalamus subcortical structures regulating wakefulness to influence behaviour, which intrinsically primarily depends on cortical activity.

Future research may consider data-driven analyses of hypothalamus voxels time series as an alternative to the parcellation approach we adopted here. This may refine the delineation of the subparts of the hypothalamus undergoing decreased or increased activity with increasing illuminance. Future research should also assess the impact of light on other subcortical structures and on the entire subcortical network to determine how illuminance modifies their crosstalk as well as their interaction with the cortex, to eventually lead to behavioural impacts. These analyses

could for instance address whether the regional changes in activity of the hypothalamus we find are upstream of the repeatedly reported impact of light illuminance on the activity of the pulvinar in the thalamus (Paparella et al., 2023 [↗](#); Vandewalle et al., 2006 [↗](#)). Although it does not receive direct dense input from ipRGCs, it is likely to indirectly mediate the biological impact of light on ongoing cognitive activity (Paparella et al., 2023 [↗](#)).

We based our rationale and part of our interpretations on ipRGC projections, which have been demonstrated in rodents to channel the NIF biological impact of light and incorporate the inputs from rods and cones with their intrinsic photosensitivity into a light signal that can impact the brain (Güler et al., 2008 [↗](#); Tri & Do, 2019 [↗](#)). Given the polychromatic nature of the light we used, classical photoreceptors and their projections to visual brain areas are, however, very likely to have directly or indirectly contributed to the modulation by light of the regional activity of the hypothalamus. Furthermore, we cannot exclude that colour and/or spectral differences between the orange and 3 blue-enriched light conditions may have contributed to our findings. Research in rodent models demonstrated that variation in the spectral composition of light was perceived by the suprachiasmatic nucleus to set circadian timing (Walmsley et al., 2015 [↗](#)). No such demonstration has, however, been reported yet for the acute impact of light on alertness, attention, cognition or affective state. Future human studies could isolate the contribution of each photoreceptor class to the impact of light on cognitive brain functions by manipulating prior light history (Chellappa et al., 2014 [↗](#)) or through the use of silent substitutions between metameric light exposures (Viénot et al., 2012 [↗](#)).

All these knowledge gaps are important to address because acting on light stands as a promising means to reduce high sleepiness and improve cognitive deficits during wakefulness as well as to facilitate sleep in the few hours preceding bedtime (Brown et al., 2022 [↗](#); Didikoglu et al., 2023 [↗](#); Gooley et al., 2011 [↗](#); Munch et al., 2016 [↗](#); Riemersma-van der Lek et al., 2008 [↗](#); Scheuermaier et al., 2018 [↗](#); Wirz-Justice et al., 2020 [↗](#)). Light therapy is also a validated means to improve mood and treat mood disorders (Glickman et al., 2006 [↗](#); Lam et al., 2016 [↗](#)). Light administration can also be considered a simple means to disturb the brain circuitry regulating sleep and wakefulness such that it can provide insights about novel means to improve their quality. For instance, if orexin and histamine were part of the mechanism through which natural light affects brain functions, their administration may be the most ecological and/or natural means to affect alertness and cognition. Likewise, as both orexin and histamine are targets for the treatment of brain disorders, our findings could suggest that light may constitute a non-pharmacological complementary intervention to compounds that are being developed to treat arousal, sleep, or cognitive dysfunction in brain disorders (Ma et al., 2018 [↗](#)). It remains, however, premature in our view to base recommendations on the therapeutic use of light based on the MRI findings gathered to date. Targeted lighting for interventions or for precise interference of subcortical circuits will require a full understanding of how light affects the brain, particularly at the subcortical level. Our findings represent an important step towards this goal, at the level of the hypothalamus.

Materials and Methods

The data used in this paper arise from a large study that is leading to several publications and part of the methods have been published previously (Beckers et al., 2023 [↗](#); Campbell, Beckers, et al., 2023 [↗](#); Paparella et al., 2023 [↗](#)). The protocol was approved by the Ethics Committee of the Faculty of Medicine at the University of Liège. Participants gave their written informed consent to take part in the study and received monetary compensation for their participation.

Participants

Thirty healthy young adults (19 women; 24.3 ± 2.9 y; **Suppl. Table S1**) were included in the analyses. Exclusion criteria were assessed through questionnaires and a semi-structured interview: history of psychiatric and neurological disorders, sleep disorders, use of psychoactive drugs or addiction; history of ophthalmic disorders or auditory impairments; colour blindness; night shift work during the last year or recent trans-meridian travel during the last 2 months; excessive caffeine (>4 caffeine units/day) or alcohol consumption (>14 alcohol units/week); medication affecting the central nervous system; smoking; pregnancy or breast-feeding (women); counter indication for MRI-scanning. All participants had to score < 18 on the 21-item Beck Anxiety Inventory (up to mild anxiety) (Beck, Epstein, et al., 1988), and <14 on the Beck Depression Inventory-II (up to mild depression) (Beck, Steer, et al., 1988), < 12 on the Epworth Sleepiness Scale (Johns, 1993 [\[1\]](#)), and < 8 on the Pittsburgh Sleep Quality Index (Buysse et al., 1989 [\[2\]](#)). Questionnaires further assessed chronotype with the Horne-Östberg questionnaire (Horne & Östberg, 1976 [\[3\]](#)) and seasonality with the Seasonal Pattern Assessment Questionnaire (Rosenthal & Bradt, 1984 [\[4\]](#)), but the latter two questionnaires were not used for the inclusion of the participants.

For each task, 4 datasets were missing or had corrupt data such that 26 participants were included in the analyses of each task (23 participants had valid datasets for both tasks). For the emotional task, two participants' data failed the MRI quality control (QC) check, and the other two participants were excluded as they did not complete the entire task. For the executive task, four of the participants' data failed the MRI QC check. **Suppl. Table S2** summarises participants' characteristics respective to each task.

Overall Protocol

Participants completed an MRI session at least one week before the experiment during which structural images of the brain were acquired and which served as habituation to the experimental conditions. Participants maintained a loose sleep-wake schedule (± 1 h from the habitual sleep/wake-up time) during the 7 days preceding the fMRI experiment to warrant similar circadian entrainment across participants and avoid excessive sleep loss while maintaining realistic real-life conditions (verified using sleep diaries and wrist actigraphy - AX3 accelerometer, Axivity, United Kingdom). Volunteers were requested to refrain from all caffeine and alcohol-containing beverages, and extreme physical activity for 3 days before participating in the fMRI acquisitions. Data acquisitions took place in Liège, Belgium, between December 2020 and May 2023.

Participants arrived at the laboratory 1.5 to 2h after habitual wake time for the fMRI scan. They were first exposed for 5 min to a bright polychromatic white light (1000 lux) and then maintained in dim light (< 10 lux) for 45 min to standardise the participant's recent light history. During this period participants were given instructions about the fMRI cognitive tasks and completed practice tasks on a luminance-controlled laptop (< 10 lux). The fMRI session consisted of participants completing three auditory cognitive tasks while alternatively maintained in darkness or exposed to light: an executive task (25 min), an emotional task (20 min) and an attentional task (15 min) [**Fig.1** [\[5\]](#) A]. The executive task was always completed first, as it was the most demanding task. The order of the following two tasks was counterbalanced. Because it included only 3 light conditions (see below) instead of 5 for the other two tasks, the attentional task was not included in the present analyses. An eye-tracking system (EyeLink 1000Plus, SR Research, Ottawa, Canada) was monitored for proper eye opening during all data acquisitions. **Light exposure.** An 8-m long MRI-compatible optic fibre (1-inch diameter, Setra Systems, MA, USA) transmitted light from a light box (SugarCUBE, Ushio America, CA, USA) to the dual end of the fibre which was attached to a stand fitted at the back of the MRI coil that allowed reproducible fixation and orientation of the optic fibre ends. The dual branches illuminated the inner walls of the head coil to ensure relatively uniform and indirect illumination of participants' eyes. A filter wheel (Spectral Products, AB300,

NM, USA) and optical fibre filters (monochromatic narrowband orange filter - 589nm; full width at half maximum: 10 nm - or a UV highpass filter - 433–1650nm) were used to create the light conditions needed for the experiment (see [Fig.1](#) [B](#) and [Suppl. Table 2](#) for in-detail light characteristics).

Illuminance and spectra could not be directly measured within the MRI scanner due to the ferromagnetic nature of measurement systems. The coil of the MRI and the light stand, together with the lighting system were therefore placed outside of the MR room to reproduce the experimental conditions of the in a completely dark room. A sensor was placed 2 cm away from the mirror of the coil that is mounted at eye level, i.e. where the eye of the first author of the paper would be positioned, to measure illuminance and spectra. The procedure was repeated 4 times for illuminance and twice for spectra and measurements were averaged. This procedure does not take into account inter-individual variation in head size and orbit shape such that the reported illuminance levels may have varied slightly across subjects. The relative differences between illuminance are, however, very unlikely to vary substantially across participants such that statistics consisting of tests for the impact of relative differences in illuminance were not affected. The detailed values reported in Supplementary Table 2 were computed combining spectra and illuminance using the excel calculator associated with a published work ([Lucas et al., 2014](#)).

Blue-enriched light illuminances were set according to the technical characteristics of the light source and to keep the overall photon flux similar to prior 3T MRI studies of our team (between $\sim 10^{12}$ and 10^{14} ph/cm²/s) ([Vandewalle et al., 2010](#); [Vandewalle, Hébert, et al., 2011](#)). The orange light was introduced as a control visual stimulation for potential secondary whole-brain analyses. For the present region of interest analyses, we discarded colour differences between the light conditions and only considered illuminance as indexed by mel EDI lux. This constitutes a limitation of our study as it does not allow attributing the findings to a particular photoreceptor class.

For the executive and emotional task, the light conditions consisted of three different illuminance of a white, blue-enriched polychromatic LED light (37, 92, 190 mel EDI lux; 6500K) and one illuminance level of monochromatic orange light (.16 mel EDI lux; 590nm full width at half maximum -FWHM: 10 nm). For the present analyses, we discarded colour differences between the light conditions and only considered illuminance as indexed by mel EDI lux, constituting a limitation of our study. In the executive task, participants were exposed to 30s to 70s (median 30s) of light blocks separated by 10s of darkness (< 0.1 lux) and the light blocks were repeated 11 times for each light condition. For the emotional task, participants were exposed to 30 to 40s (median 35s) light blocks separated by 20s of darkness (< 0.1 lux) and the light blocks were repeated five times for each light condition.

The attentional task only included a single illuminance level of the blue-enriched polychromatic LED light (92 mel EDI lux) and one illuminance level of the monochromatic orange light (.16 mel EDI lux), otherwise, the task would have been too long (> 30 min). Participants were exposed to 30s of light blocks separated by 10s of darkness (<0.1 lux). The light blocks were repeated 7 times for each light condition. As mentioned above it is not considered for the present analyses.

Cognitive Tasks

Prior work of our team showed that the n-back task and emotional task included in the present protocol were successful probes to demonstrate that light illuminance modulates cognitive activity, including within subcortical structures (though resolution did not allow precise isolation of nuclei or subparts) (e.g. ([Vandewalle et al., 2010](#); [Vandewalle, Schmidt, et al., 2007](#))). When taking the step of ultra-high-field imaging, we therefore opted for these tasks as our goal was to show that illuminance affects brain activity across cognitive domains while not testing for task-specific aspects of these domains.

Auditory cognitive tasks. The tasks were programmed with Opensesame (3.2.8) (Mathôt et al., 2012). Participants heard the auditory stimuli through MR-compatible earbuds (Sensimetrics, Malden, MA). Before starting the tasks, to ensure optimal auditory perception of task stimuli, participants set the volume through a volume check procedure. Participants used an MRI-compatible keypad to respond to task items (Current Designs, Philadelphia, PA), which was placed in the participant's dominant hand. The tasks were separated by about 5 minutes in near darkness, to recalibrate the eye tracking system and to clarify instructions about the next task to the participant.

Executive task

The task consisted of an auditory variant of the n-back task (Collette et al., 2005) with a working memory 2-back task and a control letter detection 0-back task. Participants were either asked to detect whether the current item was identical to the letter presented 2-items earlier (2-back) or whether the current item consisted of the letter “K” (0-back) or using the keypad (one button for “yes”, one button for “no”). A block design was used for this task in which each block included 15 items and lasted 30s. Task blocks were separated by 10-20s rest periods and were preceded by an auditory instruction (500 ms) indicating the type of task to be completed. Task levels were pseudo-randomised across the 4 light conditions with 3 blocks of 0-back and 4 blocks of 2-back per light condition (see Fig.1 C).

Emotional Task

The task consisted of gender discrimination of auditory vocalizations that were either pronounced with emotional or neutral prosody (Grandjean et al., 2005). Participants were asked to use the keypad to indicate what they believed the gender of the person pronouncing each token was. The gender classification was a lure task ensuring participants paid attention to the auditory stimulation. The purpose of the task was to trigger an emotional response as participants were not told that part of the stimuli was pronounced with angry prosody. The 240 auditory stimuli were pronounced by professional actors (50% women) and consisted of three meaningless words (“goster”, “niuvenci”, “figoteich”). The stimuli were expressed in either an angry or neutral prosody, which has been validated by behavioural assessments (Banse & Scherer, 1996) and in previous experiments (Grandjean et al., 2005; Sander et al., 2005; Vandewalle et al., 2010). The stimuli were also matched for the duration (750 ms) and mean acoustic energy to avoid loudness effects. During each 30 to 40-s light block, four angry prosody stimuli and four neutral prosody stimuli were presented in a pseudorandom order and delivered every 3 to 5 seconds. A total of 160 distinct voice stimuli (50% angry; 50% neutral) were distributed across the four light conditions. The darkness period separating each light block contained two angry and two neutral stimuli. A total of 80 distinct voice stimuli (50% angry; 50% neutral) were distributed across the darkness periods [see Fig.1 D].

Data acquisition

The MRI data were acquired in a 7T MAGNETOM Terra MR scanner (Siemens Healthineers, Erlangen, Germany) with a 32-channel receive and 1-channel transmit head coil (Nova Medical, Wilmington, MA, USA). Dielectric pads (Multiwave Imaging, Marseille, France) were placed between the subject's head and receiver coil to homogenize the magnetic field of Radio Frequency (RF) pulses.

Multislice T2*-weighted fMRI images were obtained with a multi-band Gradient-Recalled Echo - Echo-Planar Imaging (GRE-EPI) sequence using axial slice orientation (TR = 2340 ms, TE = 24 ms, FA = 90°, no interslice gap, in-plane FoV = 224 mm × 224 mm, matrix size = 160 × 160 × 86, voxel size = 1.4 × 1.4 × 1.4 mm³). To avoid saturation effects, the first three scans were discarded. To correct for physiological noise in the fMRI data the participants' pulse and respiration movements were recorded using a pulse oximeter and a breathing belt (Siemens Healthineers, Erlangen,

Germany). Following the fMRI acquisition a 2D GRE field mapping sequence to assess B0 magnetic field inhomogeneity with the following parameters: TR = 5.2 ms, TEs = 2.26 ms and 3.28 ms, FA = 15°, bandwidth = 737 Hz/pixel, matrix size = 96 × 128, 96 axial slices, voxel size = (2x2x2) mm³, acquisition time = 1:38 min, was applied.

For the anatomical image, a high-resolution T1-weighted image was acquired using a Magnetization-Prepared with 2 RApid Gradient Echoes (MP2RAGE) sequence: TR = 4300 ms, TE = 1.98 ms, FA = 5°/6°, TI = 940ms/2830 ms, bandwidth = 240 Hz, matrix size = 256x256, 224 axial slices, acceleration factor = 3, voxel size = (0.75x0.75x0.75) mm³.

Data Processing

For the MP2RAGE images, the background noise was removed using an extension (extension: <https://github.com/benoitberanger/mp2rage>) of Statistical Parametric Mapping 12 (SPM12; <https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>) under Matlab R2019 (MathWorks, Natick, Massachusetts) (O'Brien et al., 2014). Then the images were reoriented using the 'spm_auto_reorient' function (https://github.com/CyclotronResearchCentre/spm_auto_reorient) and corrected for intensity non-uniformity using the bias correction method implemented in the SPM12 "unified segmentation" tool (Ashburner & Friston, 2005). To ensure optimal co-registration, brain extraction was done using SynthStrip (Hoopes et al., 2022) in Freesurfer (<http://surfer.nmr.mgh.harvard.edu/>). The brain-extracted T1-images were used to create a T1-weighted group template using Advanced Normalization Tools (ANTs, <http://stnava.github.io/ANTs/>) prior to normalization to the Montreal Neurological Institute (MNI) space using ANTs (1mm³ voxel; MNI 152 template). The hypothalamus of each participant was segmented within 1mm³ MNI 152 template into 5 subparts - inferior anterior, superior anterior, inferior tubular, superior tubular, posterior [cf. **Fig.2 A**] using an automatic computational approach (Billot et al., 2020).

For the EPI images, auto reorientation was applied on the images first. Then, voxel-displacement maps were computed from the phase and magnitude images associated with B0 map acquisition (taken right after the task), using the SPM fieldmap toolbox. To correct for head motion and static and dynamic susceptibility-induced variance, the "Realign & Unwarp" of SPM12 was then applied to the EPI images. The realigned and distortion-corrected EPI images then underwent brain extraction using the SynthStrip and then the final images were smoothed with a Gaussian kernel characterized by a FWHM = 3mm. The first level analyses were performed in the native space to prevent any possible error that may be caused by co-registration.

Statistical analyses

The whole-brain univariate analyses consisted of a general linear model (GLM) computed with SPM12. For the executive task, task blocks and light blocks were modelled as block functions. For the emotional task, the auditory stimuli were modelled as stick functions. For both tasks, a high-pass filter with a 256s cut-off was applied to remove low-frequency drifts. For both tasks, stick or block functions were convolved with the canonical hemodynamic response function. Movement and physiological parameters (cardiac, and respiration), which were computed with the PhysIO Toolbox (Translational Neuromodeling Unit, ETH Zurich, Switzerland), were included as covariates of no interest (Kasper et al., 2017).

Two separate analyses were completed. In the main analyses, we sought to test whether brain responses during the tasks were modulated by overall changes in illuminance level. The regressors of task blocks or events were accompanied by a single parametric modulation regressor corresponding to the light melanopic illuminance level (.16, 37, 92, 190 mel EDI). The contrasts of interest consisted of the main effects of the parametric modulation. In the subsequent post hoc analysis, we estimated the responses to the stimuli under each light condition. Separate regressors modelled each task's block or event type under each light condition (0, 0.16, 37, 92, 190 mel EDI). The contrasts of interest consisted of the main effects of each regressor.

The output masks of the segmentation procedure we used to extract regression betas associated with each of the hypothalamus subparts using the REX Toolbox (<https://web.mit.edu/swg/software.htm>) (Duff et al., 2007). Betas were averaged (mean) within each subpart and then across the homologous subparts of each hemisphere. In the main analyses this yielded 1 activity estimate per stimulus type and per hypothalamus subpart (i.e. 10 per individual), while in the subsequent analyses, we obtained 5 activity estimates per stimulus type and per subpart (50 per individual).

For visualization of whole-brain results over the entire sample, all statistical maps obtained from the first level analysis were first transferred to the group template space and then the MNI space (1x1x1mm³ image resolution). All the registration steps were performed with ANTs. The visualisation was focused on the hypothalamus regions to assess whether increasing illuminance resulted in a local increase and/ or decrease of beta estimates within the hypothalamus or whether beta estimates were mainly influenced by a relatively unspecific and widespread increase in BOLD signal surrounding the hypothalamus.

Statistical analyses of the activity of the hypothalamus subparts were performed in SAS 9.4 (SAS Institute, NC, USA). Analyses consisted of Generalised Linear Mixed Models (GLMM) with the subject as a random factor (intercept and slope) and were adjusted for the dependent variable distribution. As the main statistical analysis was completed for each task, the significance threshold was corrected for multiple comparisons and was set at $p < 0.025$. Direct post hoc of the main analyses were corrected for multiple comparisons using a Tukey adjustment. The subsequent more detailed analyses were considered as post hoc that were not corrected for multiple comparisons ($p < 0.05$). To detect outlier values within the data sets, Cook's distance > 1 was used for exclusion. No outliers were detected for activity estimates of both tasks, while four outlier values were removed from the analyses of the 2-back and 0-back performance.

The main analyses included the activity estimates modulated by light illuminance as a dependent variable and the hypothalamus subpart and stimulus type (2-back/0-back - neutral/emotional) as repeated measures (autoregressive (1) correlation), together with age, sex and BMI as covariates. The second set of post hoc GLMM analyses included the activity estimates of the hypothalamus subparts as the dependent variable and hypothalamus subpart, stimulus type and illuminance (0, 0.16, 37, 92, 190 mel EDI lux) as the repeated measures (autoregressive (1) correlation), together with age, sex, and BMI as covariates and interaction term between illuminance and hypothalamus subpart. The final set of analyses included performance metrics as dependent variables (accuracy to the 2-back or 0-back task - as percentage of correct responses; reaction time - ms - to emotional or neutral stimuli during the emotional task) and included the same repeated measures and covariates as in the preceding set as well as activity of the relevant hypothalamus subpart.

Optimal sensitivity and power analyses in GLMMs remain under investigation (e.g. (Kain et al., 2015)). We nevertheless computed a prior sensitivity analysis to get an indication of the minimum detectable effect size in our main analyses given our sample size. According to G*Power 3 (version 3.1.9.4) (Erdfeider et al., 2009), taking into account a power of 0.8, an error rate α of 0.025 (correcting for 2 tasks), and a sample of 26 allowed us to detect large effect sizes $r > 0.54$ (two-sided; absolute values; CI: 0.19–0.77; $R^2 > 0.29$, R^2 CI: 0.04–0.59) within a multiple linear regression framework including one tested predictor (illuminance effect) and three covariates (age, sex and BMI).

Data and code availability statement

The processed data and analysis scripts supporting the results included in this manuscript are publicly available via the following open repository: <https://gitlab.uliege.be/CyclotronResearchCentre/Public/xxxx> (the repository will be created with the version of records).

Author Contributions

I.C., R. S. and G.V. designed the research. I.C., R.S., J.F.B.A., E.B. and I.P. acquired the data. A.B., E.K., N.M., J.R., M.Z., P.T., F.C., S.S., C.P., and L.L. provided valuable insights while acquiring, interpreting, and discussing the data. I.C. and R.S. analysed the data supervised by G.V. I.C. and G.V. wrote the paper. All authors edited and approved the final version of the manuscript.

Competing Interest Statement

The authors declare no conflict of interest.

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Editors

Reviewing Editor

Birte Forstmann

University of Amsterdam, Amsterdam, Netherlands

Senior Editor

Michael Frank

Brown University, Providence, United States of America

Reviewer #1 (Public Review):

Summary:

Campbell et al investigated the effects of light on the human brain, in particular the subcortical part hypothalamus during auditory cognitive tasks. The mechanisms and neuronal circuits underlying light effects in non-image forming responses are so far mostly studied in rodents but are not easily translated in humans. Therefore, this is a fundamental study aiming to establish the impact light illuminance has on the subcortical structures using the high-resolution 7T fMRI. The authors found that parts of the hypothalamus are differently responding to illuminance. In particular, they found that the activity of the posterior hypothalamus increases while the activity of the anterior and ventral parts of the hypothalamus decreases under high illuminance. The authors also report that the performance of the 2-back executive task was significantly better in higher illuminance conditions. However, it seems that the activity of the posterior hypothalamus subpart is negatively related to the performance of the executive task, implying that it is unlikely that this part of the hypothalamus is directly involved in the positive impact of light on performance observed. Interestingly, the activity of the posterior hypothalamus was, however, associated with an increased behavioural response to emotional stimuli. This suggests that the role of this posterior part of the hypothalamus is not as simple regarding light effects on cognitive and emotional responses. This study is a fundamental step towards our better understanding of the mechanisms underlying light effects on cognition and consequently optimising lighting standards.

Strengths:

While it is still impossible to distinguish individual hypothalamic nuclei, even with the high-resolution fMRI, the authors split the hypothalamus into five areas encompassing five groups of hypothalamic nuclei. This allowed them to reveal that different parts of the hypothalamus respond differently to an increase in illuminance. They found that higher illuminance increased the activity of the posterior part of the hypothalamus encompassing the MB and parts of the LH and TMN, while decreasing the activity of the anterior parts encompassing the SCN and another part of TMN. These findings are somewhat in line with studies in animals. It was shown that parts of the hypothalamus such as SCN, LH, and PVN receive direct retinal input in particular from ipRGCs. Also, acute chemogenetic activation of ipRGCs was shown to induce activation of LH and also increased arousal in mice.

Weaknesses:

While the light characteristics are well documented and EDI calculated for all of the photoreceptors, it is not very clear why these irradiances and spectra were chosen. It would be helpful if the authors explained the logic behind the four chosen light conditions tested. Also, the lights chosen have cone-opic EDI values in a high correlation with the melanopic EDI, therefore we can't distinguish if the effects seen here are driven by melanopsin and/or other photoreceptors. In order to provide a more mechanistic insight into the light-driven effects on cognition ideally one would use silent substitution approach to distinguish between different photoreceptors. This may be something to consider when designing the follow-up studies.

<https://doi.org/10.7554/eLife.96576.2.sa3>

Reviewer #2 (Public Review):

Summary

The interplay between environmental factors and cognitive performance has been a focal point of neuroscientific research, with illuminance emerging as a significant variable of interest. The hypothalamus, a brain region integral to regulating circadian rhythms, sleep, and alertness, has been posited to mediate the effects of light exposure on cognitive functions. Previous studies have highlighted the role of the hypothalamus in orchestrating bodily responses to light, implicating specific neural pathways such as the orexin and histamine systems, which are crucial for maintaining wakefulness and processing environmental cues. Despite advancements in our understanding, the specific mechanisms through which varying levels of light exposure influence hypothalamic activity and, in turn, cognitive performance, remain inadequately explored. This gap in knowledge underscores the need for high-resolution investigations that can dissect the nuanced impacts of illuminance on different hypothalamic regions. Utilizing state-of-the-art 7 Tesla functional magnetic resonance imaging (fMRI), the present study aims to elucidate the differential effects of light on hypothalamic dynamics and establish a link between regional hypothalamic activity and cognitive outcomes in healthy young adults. By shedding light on these complex interactions, this research endeavours to contribute to the foundational knowledge necessary for developing innovative therapeutic strategies aimed at enhancing cognitive function through environmental modulation.

Strengths:

- (1) Considerable Sample Size and Detailed Analysis: The study leverages a robust sample size and conducts a thorough analysis of hypothalamic dynamics, which enhances the reliability and depth of the findings.
- (2) Use of High-Resolution Imaging: Utilizing 7 Tesla fMRI to analyze brain activity during cognitive tasks offers high-resolution insights into the differential effects of illuminance on hypothalamic activity, showcasing the methodological rigour of the study.
- (3) Novel Insights into Illuminance Effects: The manuscript reveals new understandings of how different regions of the hypothalamus respond to varying illuminance levels, contributing valuable knowledge to the field.
- (4) Exploration of Potential Therapeutic Applications: Discussing the potential therapeutic applications of light modulation based on the findings suggests practical implications and future research directions.

The current version of the manuscript addresses previous weaknesses, including details about the illuminance levels, light spectral characteristics used in the MRI study, and light patterns during behavioural tasks. The authors effectively tackle open questions in the field

and provide solid evidence that enhances our understanding of the mechanisms underlying the effects of light on cognition.

<https://doi.org/10.7554/eLife.96576.2.sa2>

Reviewer #3 (Public Review):

Summary:

Campbell and colleagues use a combination of high-resolution fMRI, cognitive tasks and different intensities of light illumination to test the hypothesis that the intensity of illumination differentially impacts hypothalamic substructures that, in turn, promote alterations in arousal that affect cognitive and affective performance. The authors find evidence in support of a posterior-to-anterior gradient of increased blood flow in the hypothalamus during task performance that they later relate to performance on two different tasks. The results provide an enticing link between light levels, hypothalamic activity and cognitive/affective function, however clarification of some methodological choices will help to improve confidence in the findings.

Strengths:

* The authors' focus on the hypothalamus and its relationship to light intensity is an important and understudied question in neuroscience.

Weaknesses:

* I found it challenging to relate the authors hypotheses, which I found to be quite compelling, to the apparatus used to test the hypotheses - namely, the use of orange light vs. different light intensities; and the specific choice of the executive and emotional tasks, which differed in key features (e.g., block-related vs. event-related designs) that were orthogonal to the psychological constructs being challenged in each task.

* Given the small size of the hypothalamus and the irregular size of the hypothalamic parcels, I wondered whether a more data-driven examination of the hypothalamic time series would have provided a more parsimonious test of their hypothesis.

<https://doi.org/10.7554/eLife.96576.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

Campbell et al investigated the effects of light on the human brain, in particular the subcortical part of the hypothalamus during auditory cognitive tasks. The mechanisms and neuronal circuits underlying light effects in non-image forming responses are so far mostly studied in rodents but are not easily translated in humans. Therefore, this is a fundamental study aiming to establish the impact light illuminance has on the subcortical structures using the high-resolution 7T fMRI. The authors found that parts of the hypothalamus are differently responding to illuminance. In particular, they found that the activity of the posterior hypothalamus increases while the activity of the anterior and ventral parts of the hypothalamus decreases under high illuminance. The authors

also report that the performance of the 2-back executive task was significantly better in higher illuminance conditions. However, it seems that the activity of the posterior hypothalamus subpart is negatively related to the performance of the executive task, implying that it is unlikely that this part of the hypothalamus is directly involved in the positive impact of light on performance observed. Interestingly, the activity of the posterior hypothalamus was, however, associated with an increased behavioural response to emotional stimuli. This suggests that the role of this posterior part of the hypothalamus is not as simple regarding light effects on cognitive and emotional responses. This study is a fundamental step towards our better understanding of the mechanisms underlying light effects on cognition and consequently optimising lighting standards.

Strengths:

While it is still impossible to distinguish individual hypothalamic nuclei, even with the highresolution fMRI, the authors split the hypothalamus into five areas encompassing five groups of hypothalamic nuclei. This allowed them to reveal that different parts of the hypothalamus respond differently to an increase in illuminance. They found that higher illuminance increased the activity of the posterior part of the hypothalamus encompassing the MB and parts of the LH and TMN, while decreasing the activity of the anterior parts encompassing the SCN and another part of TMN. These findings are somewhat in line with studies in animals. It was shown that parts of the hypothalamus such as SCN, LH, and PVN receive direct retinal input in particular from ipRGCs. Also, acute chemogenetic activation of ipRGCs was shown to induce activation of LH and also increased arousal in mice.

Weaknesses:

While the light characteristics are well documented and EDI calculated for all of the photoreceptors, it is not very clear why these irradiances and spectra were chosen. It would be helpful if the authors explained the logic behind the four chosen light conditions tested. Also, the lights chosen have cone-opic EDI values in a high correlation with the melanopic EDI, therefore we can't distinguish if the effects seen here are driven by melanopsin and/or other photoreceptors. In order to provide a more mechanistic insight into the light-driven effects on cognition ideally one would use a silent substitution approach to distinguish between different photoreceptors. This may be something to consider when designing the follow-up studies.

Reviewer #1 (Recommendations For The Authors):

(1) As suggested in the public review more information regarding the reasons behind the chosen light condition is needed.

While the light characteristics are well documented and EDI calculated for all of the photoreceptors, it is not very clear why these irradiances and spectra were chosen. It would be helpful if the authors explained the logic behind the four chosen light conditions tested. Also, the lights chosen have cone-opic EDI values in a high correlation with the melanopic EDI, therefore we can't distinguish if the effects seen here are driven by melanopsin or cone opsins. In order to provide a more mechanistic insight into the light-driven effects on cognition ideally one would use a silent substitution approach to distinguish between different photoreceptors.

(2) In support of this work, it was shown in mice that acute activation of ipRGCs using chemogenetics induces c-fos in some of the hypothalamic brain areas discussed here including LH (Milosavljevic et al, 2016 Curr Biol). Another study to consider including in the discussion is by Sonoda et al 2020 Science, in which the authors showed that a subset of ipRGCs release GABA.

(3) Figure 1 looks squashed, especially the axes. Also, Figure 2 looks somewhat blurry. I would suggest that the authors edit the figures to correct this.

We thank the reviewer for their positive comments and agree with the weaknesses they pointed out.

(1) The explanation regarding the choice of the illuminance is now included in the revised manuscript (PAGE 17): “Blue-enriched light illuminances were set according to the technical characteristics of the light source and to keep the overall photon flux similar to prior 3T MRI studies of our team (between ~1012 and 1014 ph/cm²/s) (Vandewalle et al., 2010, 2011). The orange light was introduced as a control visual stimulation for potential secondary whole-brain analyses. For the present region of interest analyses, we discarded colour differences between the light conditions and only considered illuminance as indexed by mel EDI lux. This constitutes a limitation of our study as it does not allow attributing the findings to a particular photoreceptor class.”

The revised discussion makes clear that these choices limit the interpretation about the photoreceptors involved (PAGES 12-13): “We based our rationale and part of our interpretations on ipRGC projections, which have been demonstrated in rodents to channel the NIF biological impact of light and incorporate the inputs from rods and cones with their intrinsic photosensitivity into a light signal that can impact the brain (Güler et al., 2008; Tri & Do, 2019). Given the polychromatic nature of the light we used, classical photoreceptors and their projections to visual brain areas are, however, very likely to have directly or indirectly contributed to the modulation by light of the regional activity of the hypothalamus.”

The discussion also points out the promises of silent substitution (PAGE 13): “Future human studies could isolate the contribution of each photoreceptor class to the impact of light on cognitive brain functions by manipulating prior light history (Chellappa et al., 2014) or through the use of silent substitutions between metameric light exposures (Viénot et al., 2012)”.

(2) We now refer to the studies by Milosavljevic et al. and Sonoda et al.

PAGE 9: “Our data may therefore be compatible with an increase in orexin release by the LH with increasing illuminance. In line with this assumption, chemoactivation of ipRGCs lead to increase c-fos production, a marker of cellular activation, over several nuclei of the hypothalamus, including the lateral hypothalamus (Milosavljevic et al., 2016). If this initial effect of light we observe over the posterior part of the hypothalamus was maintained over a longer period of exposure, this would stimulate cognition and maintain or increase alertness (Campbell et al., 2023) and may also be part of the mechanisms through which daytime light increases the amplitude in circadian variations of several physiological features (BanoOtalora et al., 2021; Dijk et al., 2012).”

PAGE 10: “Chemoactivation of ipRGCs in rodents led to an increase activity of the SCN, over the inferior anterior hypothalamus, but had no impact on the activity of the VLPO, over the superior anterior hypothalamus (Milosavljevic et al., 2016). How our findings fit with these fine-grained observations and whether there are species-specific differences in the responses to light over the different part of the hypothalamus remains to be established.”

PAGE 10: “In terms of chemical communication, these changes in activity could be the results of an inhibitory signal from a subclass of ipRGCs, potentially through the release aminobutyric acid (GABA), as a rodent study found that a subset of ipRGCs release GABA at brain targets including the SCN (and intergeniculate leaflet and ventral lateral geniculate nucleus), leading to a reduction in the ability of light to affect pupil size and circadian photoentrainment (Sonoda et al., 2020). Whatever the signalling of ipRGC, our finding over the anterior hypothalamus could correspond to a modification of GABA signalling of the SCN

which has been reported to have excitatory properties, such that the BOLD signal changes we report may correspond to a reduction in excitation arising in part from the SCN (Albers et al., 2017).”

(3) Figures 1 and 2 were modified. We hope their quality is now satisfactory. We are willing to provide separate figures prior to publication of the Version of Record.

Reviewer #2 (Public Review):

Summary

The interplay between environmental factors and cognitive performance has been a focal point of neuroscientific research, with illuminance emerging as a significant variable of interest. The hypothalamus, a brain region integral to regulating circadian rhythms, sleep, and alertness, has been posited to mediate the effects of light exposure on cognitive functions. Previous studies have illuminated the role of the hypothalamus in orchestrating bodily responses to light, implicating specific neural pathways such as the orexin and histamine systems, which are crucial for maintaining wakefulness and processing environmental cues. Despite advancements in our understanding, the specific mechanisms through which varying levels of light exposure influence hypothalamic activity and, in turn, cognitive performance, remain inadequately explored. This gap in knowledge underscores the need for high-resolution investigations that can dissect the nuanced impacts of illuminance on different hypothalamic regions. Utilizing state-of-the-art 7 Tesla functional magnetic resonance imaging (fMRI), the present study aims to elucidate the differential effects of light on the hypothalamic dynamics and establish a link between regional hypothalamic activity and cognitive outcomes in healthy young adults. By shedding light on these complex interactions, this research endeavours to contribute to the foundational knowledge necessary for developing innovative therapeutic strategies aimed at enhancing cognitive function through environmental modulation.

Strengths:

(1) Considerable Sample Size and Detailed Analysis: The study leverages a robust sample size and conducts a thorough analysis of hypothalamic dynamics, which enhances the reliability and depth of the findings.

(2) Use of High-Resolution Imaging: Utilizing 7 Tesla fMRI to analyze brain activity during cognitive tasks offers high-resolution insights into the differential effects of illuminance on hypothalamic activity, showcasing the methodological rigor of the study.

(3) Novel Insights into Illuminance Effects: The manuscript reveals new understandings of how different regions of the hypothalamus respond to varying illuminance levels, contributing valuable knowledge to the field.

(4) Exploration of Potential Therapeutic Applications: Discussing the potential therapeutic applications of light modulation based on the findings suggests practical implications and future research directions.

Weaknesses:

(1) Foundation for Claims about Orexin and Histamine Systems: The manuscript needs to provide a clearer theoretical or empirical foundation for claims regarding the impact of light on the orexin and histamine systems in the abstract.

(2) Inclusion of Cortical Correlates: While focused on the hypothalamus, the manuscript may benefit from discussing the role of cortical activation in cognitive performance,

suggesting an opportunity to expand the scope of the manuscript.

(3) Details of Light Exposure Control: More detailed information about how light exposure was controlled and standardized is needed to ensure the replicability and validity of the experimental conditions.

(4) Rationale Behind Different Exposure Protocols: To clarify methodological choices, the manuscript should include more in-depth reasoning behind using different protocols of light exposure for executive and emotional tasks.

Reviewer #2 (Recommendations For The Authors):

Attention to English language precision and correction of typographical errors, such as "hypothalamic nuclei" instead of "hypothalamus nuclei," is necessary for enhancing the manuscript.

We thank the reviewer for recognising the interest and strength of our study.

(1) As detailed in the discussion, we do believe orexin and histamine are excellent candidates for mediating the results we report. As also pointing out, however, we are in no position to know which neurons, nuclei, neurotransmitter and neuromodulator underlie the results. The last sentence of the abstract (PAGE 2) was therefore removed as we agree the statement was too strong. We carefully reconsider the discussion and believe that no such overstatement was present.

(2) Hypothalamus nuclei are connected to multiple cortical (and subcortical) structures. The relevance of these projections will vary with the cognitive task considered. In addition, we have not yet considered the cortex in our analyses such that truly integrating cortical structures appears premature.

We nevertheless added the following short statement (PAGE 11): "Subcortical structures, and particularly those receiving direct retinal projections, including those of the hypothalamus, are likely to receive light illuminance signal first before passing on the light modulation to the cortical regions involved in the ongoing cognitive process (Campbell et al., 2023)."

(3) We now include the following as part of the method section (PAGES 16-17): "Illuminance and spectra could not be directly measured within the MRI scanner due to the ferromagnetic nature of measurement systems. The coil of the MRI and the light stand, together with the lighting system were therefore placed outside of the MR room to reproduce the experimental conditions of the in a completely dark room. A sensor was placed 2 cm away from the mirror of the coil that is mounted at eye level, i.e. where the eye of the first author of the paper would be positioned, to measure illuminance and spectra. The procedure was repeated 4 times for illuminance and twice for spectra and measurements were averaged. This procedure does not take into account interindividual variation in head size and orbit shape such that the reported illuminance levels may have varied slightly across subjects. The relative differences between illuminance are, however, very unlikely to vary substantially across participants such that statistics consisting of tests for the impact of relative differences in illuminance were not affected. The detailed values reported in Supplementary Table 2 were computed combining spectra and illuminance using the excel calculator associated with a published work (Lucas et al., 2014)."

(4) The explanation regarding the choice of the illuminance is now included in the revised manuscript (PAGE 17): "Blue-enriched light illuminances were set according to the technical characteristics of the light source and to keep the overall photon flux similar to prior 3T MRI studies of our team (between ~1012 and 1014 ph/cm²/s) (Vandewalle et al., 2010, 2011). The orange light was introduced as a control visual stimulation for potential secondary whole-brain analyses. For the present region of interest analyses, we discarded colour differences

between the light conditions and only considered illuminance as indexed by mel EDI lux. This constitutes a limitation of our study as it does not allow attributing the findings to a particular photoreceptor class.”

(5) The manuscript was thoroughly rechecked, and we hope to have spotted all typos and language errors.

Reviewer #3 (Public Review):

Summary:

Campbell and colleagues use a combination of high-resolution fMRI, cognitive tasks, and different intensities of light illumination to test the hypothesis that the intensity of illumination differentially impacts hypothalamic substructures that, in turn, promote alterations in arousal that affect cognitive and affective performance. The authors find evidence in support of a posterior-to-anterior gradient of increased blood flow in the hypothalamus during task performance that they later relate to performance on two different tasks. The results provide an enticing link between light levels, hypothalamic activity, and cognitive/affective function, however, clarification of some methodological choices will help to improve confidence in the findings.

Strengths:

** The authors' focus on the hypothalamus and its relationship to light intensity is an important and understudied question in neuroscience.*

Weaknesses:

(1) I found it challenging to relate the authors' hypotheses, which I found to be quite compelling, to the apparatus used to test the hypotheses - namely, the use of orange light vs. different light intensities; and the specific choice of the executive and emotional tasks, which differed in key features (e.g., block-related vs. event-related designs) that were orthogonal to the psychological constructs being challenged in each task.

(4) Given the small size of the hypothalamus and the irregular size of the hypothalamic parcels, I wondered whether a more data-driven examination of the hypothalamic time series would have provided a more parsimonious test of their hypothesis.

Reviewer #3 (Recommendations For The Authors):

(1) The authors may wish to explain the importance of the orange light condition in the early section of the results -- i.e., when they first present the task structure. As it stands, I don't have a good appreciation of why the orange light was included -- was it a control condition? And if the differences between the light conditions (e.g., the narrow- vs. wide-band of light) were indeed ignored by focussing on the illuminance levels, are there any potential issues that the authors could then mitigate against with further experiments/analyses?

(2) Are there other explanations for why illuminance levels might improve cognitive performance? For instance, the capacity to more easily perceive the stimuli in an experiment could plausibly make it easier to complete a given task. If this is the case, can the authors conceptualise a way to rule out this hypothesis?

(3) Did the authors control for the differences in the number of voxels in each hypothalamic subregion? Or perhaps consider estimating the variance across voxels within the larger parcels, to determine whether the mean time series was comparable to the time series of the smaller parcels?

(4) An alternative strategy that would mitigate against the differences in the size of hypothalamic parcels would be to conduct analyses on the hypothalamus without parcellation, but instead using dimensionality reduction techniques to observe the natural spread of responses across the hypothalamus. From the authors' results, my intuition is that these analyses will lead to similar conclusions, albeit without any of the potential issues with respect to differently-sized parcels.

We thank the reviewer for acknowledging the originality and interest of our study. We agree that some methodological choices needed more explanation. We will address the weaknesses they pointed out as follows:

(1) The explanation regarding the choice of the illuminance is now included in the revised manuscript (PAGE 17): “Blue-enriched light illuminances were set according to the technical characteristics of the light source and to keep the overall photon flux similar to prior 3T MRI studies of our team (between ~1012 and 1014 ph/cm²/s) (Vandewalle et al., 2010, 2011). The orange light was introduced as a control visual stimulation for potential secondary whole-brain analyses. For the present region of interest analyses, we discarded colour differences between the light conditions and only considered illuminance as indexed by mel EDI lux. This constitutes a limitation of our study as it does not allow attributing the findings to a particular photoreceptor class.”

The revised discussion makes clear that these choices limit the interpretation about the photoreceptors involved (PAGE 12-13): “We based our rationale and part of our interpretations on ipRGC projections, which have been demonstrated in rodents to channel the NIF biological impact of light and incorporate the inputs from rods and cones with their intrinsic photosensitivity into a light signal that can impact the brain (Güler et al., 2008; Tri & Do, 2019). Given the polychromatic nature of the light we used, classical photoreceptors and their projections to visual brain areas are, however, very likely to have directly or indirectly contributed to the modulation by light of the regional activity of the hypothalamus.”

We further mention that (PAGE 13): “Furthermore, we cannot exclude that colour and/or spectral differences between the orange and 3 blue-enriched light conditions may have contributed to our findings. Research in rodent model demonstrated that variation in the spectral composition of light was perceived by the suprachiasmatic nucleus to set circadian timing (Walmsley et al., 2015). No such demonstration has, however, been reported yet for the acute impact of light on alertness, attention, cognition or affective state.”

Regarding the choice of tasks, we added the following the method section (PAGE 18): “Prior work of our team showed that the n-back task and emotional task included in the present protocol were successful probes to demonstrate that light illuminance modulates cognitive activity, including within subcortical structures (though resolution did not allow precise isolation of nuclei or subparts) (e.g. (Vandewalle et al., 2007, 2010)). When taking the step of ultra-high-field imaging, we therefore opted for these tasks as our goal was to show that illuminance affects brain activity across cognitive domains while not testing for task-specific aspects of these domains.”

We further added to the discussion (PAGE 8): “The pattern of light-induced changes was consistent across an executive and an emotional task which consisted of block and an event-related fMRI design, respectively. This suggests that a robust anterior-posterior gradient of activity modulation by illuminance is present in hypothalamus across cognitive domains.”

(2) We are unsure what the reviewer refers to when he states that the experiment could make it easier to perceive a stimulus. Aside from the fact that illuminance can increase alertness and attention such that a stimulus may be better or more easily perceived/processed, we do not see how blocks of ambient light, i.e. a long-lasting visual stimulus, may render auditory

stimulation (letters or pseudo-words in the present) easier to perceive. To our knowledge multimodal or cross-modal integration has been robustly demonstrated for short visual/auditory cues that would precede or accompany auditory/visual stimulation.

We are willing to clarify this issue in the text if we receive additional explanation from the reviewer.

(3) We added subpart size as covariate in the analyses (instead of subpart number) and it did not affect the output of the statistical analyses (Author response table 1).

For completeness, we further computed standard deviation of the activity estimates of the voxels within each parcel for the main analysis of the n-back tasks and found a main effect of subpart (Author response table 2) indicating that the variability of the estimates varied across subparts. Post hoc contrast and the display included in Author response image1 show however that the difference were not related to subpart size per se. It is in fact the largest subpart (subpart 4) that shows the largest variability while one of the smallest subpart (subpart 2) shows the lowest variability. Though it may have contributed, it is therefore unlikely to explain our findings. We consider the analyses reported in (Author response table 1 and 2 and (Author response image 1 as very technical and did not include it in the supplementary material for conciseness. If the reviewer judges it essential, we can reconsider our decision.

While computing these analyses, we realized that there were errors in the table 1 reporting the statistical outcomes of the main analyses of the emotional task. The main statistical outputs remain the same except for a nominal main effect of the task (emotional vs. neutral) and the fact that post hoc show a consistent difference between the posterior subpart (subpart 3) and all the other subparts, rather than all the other subparts except for the difference with superior tubular hypothalamus subpart: p-corrected = 0.09. We apologise for this slight error and were unable to isolate its origin. It does not modify the rest of the analyses (which were also rechecked) and the interpretations.

Author response table 1.

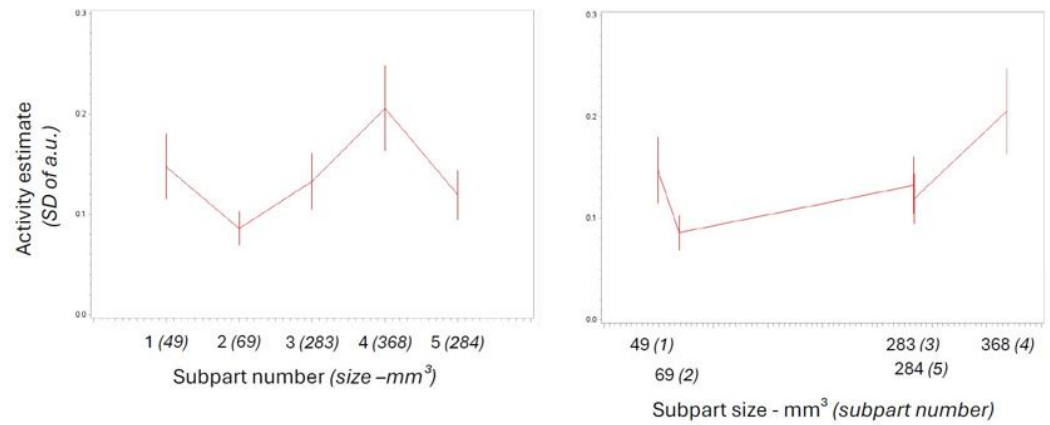
Recomputations of the main GLMMs using subpart sizes rather than subpart numbers as covariate of interest.

Executive task			
Main GLMM			
Effect	F value (df)	P value*	Partial R ²
Subpart size	4.36 (4,200)	0.0021	0.08
Task	0.74 (1,25)	0.4	
Subpart size x task type	0.68 (4,200)	0.61	
Age	0.33 (1,22)	0.57	
BMI	0.59 (1,22)	0.45	
Sex	0.01 (1,22)	0.91	
Emotional task			
Main GLMM			
Effect	F Value	P value*	Partial R ²
Subpart size	9.64 (4,193)	<.0001	0.17
Task	4.55 (1,25)	0.043	0.15
Subpart size x stimulus type	0.41 (4,193)	0.8	
Age	0.4 (1,22)	0.53	
BMI	0.02 (1,22)	0.89	
Sex	1.37 (1,22)	0.25	

Author response image 1.

Activity estimate variability per hypothalamus subpart and subpart size.

Activity estimate variability per hypothalamus subpart and subpart size



Author response table 2.

Difference in activity estimate standard deviation between hypothalamus subparts during the n-back task.

Subpart number (GLMM)			
Effect	F value (df)	P value*	Partial R ²
Hypothalamus subparts	32.3 (4,260)	<0.0001	0.33
Task	<0.001 (1,23)	0.97	
Hypothalamus subparts x task type	0.03 (4,260)	0.99	
Post hoc			
subpart	T value	p	P corrected
1 vs. 2	5.24	<.0001	<.0001
1 vs. 3	0.30	0.77	0.99
1 vs. 4	1.47	0.14	0.58
1 vs. 5	-5.92	<.0001	<.0001
2 vs. 3	-4.95	<.0001	<.0001
2 vs. 4	-3.77	0.0002	0.0019
2 vs. 5	-11.17	<.0001	<.0001
3 vs. 4	1.18	0.24	0.77
3 vs. 5	-6.22	<.0001	<.0001
4 vs. 5	-7.39	<.0001	<.0001

Outputs of the generalized linear mixed model (GLMM) with subject as the random factor (intercept and slope), and task and subpart as repeated measures (ar(1) autocorrelation).

* The corrected p-value for multiple comparisons over 2 tests is $p < 0.025$.

Refer to Fig.2A for correspondence of subpart numbers

The text referring to Table 1 was modified accordingly (PAGE 5): “A nominal main effect of the task was detected for the emotional task [$p = 0.049$; Table 1] but not for the n-back task. For both tasks, there was no significant main effect for any of the other covariates and post hoc analyses showed that the index of the illuminance impact was consistently different in the posterior hypothalamus subpart compared to the other subparts [$p_{\text{corrected}} \leq 0.05$]”.

(4) We agree that a data driven approach could have constituted an alternative means to tests our hypothesis. We opted for an approach that we mastered best, while still allowing to conclusively test for regional differences in activity across the hypothalamus. Examination of time series of the very same data we used will mainly confirm the results of our analyses – an

anterior-posterior gradient in the impact of illuminance - while it may yield slight differences in the borders of the subparts of the hypothalamus undergoing decreased or increased activity with increasing illuminance. While the suggested approach may have been envisaged if we had been facing negative results (i.e. no differences between subparts, potentially because subparts would not reflect functional differences in response to illuminance change), it would constitute a circular confirmation of our main findings (i.e. using the same data). While we truly appreciate the suggestion, we do not consider that it would constitute a more parsimonious test of our hypothesis, now that we successfully applied GLM/parcellation and GLMM approaches.

We added the following statement to the discussion to take this comment into account (PAGE 12): “Future research may consider data-driven analyses of hypothalamus voxels time series as an alternative to the parcellation approach we adopted here. This may refine the delineation of the subparts of the hypothalamus undergoing decreased or increased activity with increasing illuminance.”

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